Forces in Vector Components

Vectors usually have magnitude + direction

Forces can be written as vectors and therefore can be broken up into vector components

Vector components are parts of a vector wrt cartesian coordinates (x, y, z)

We can use the fact that sum of vector components $(F_x, F_y, F_z) = 0$ when system @ EQ to find unknown

$\vec{F} = 5N$ @ $53.13°$ from ground

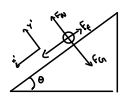
$\vec{F}_x = 5\sin(53.13°)N = 3\hat{i}\ N$

$\vec{F}_y = 5\cos(53.13°)N = 4\hat{j}\ N$

FBD & Particle Diagrams

If possible converting reference from coordinate plane can help (PHYS192 wedge problem)

Forces labeled w/ subscripts are always worked from start to end (F_{AB} goes from A to B)



$F_W = F_{Gy} = F_R$

Moments of Rotation

Rigid bodies rotate unlike static particles which needs to be accounted for

Therefore for an object @ EQ, $\Sigma F_x, \Sigma F_y, \Sigma F_z = 0$ but also $\Sigma M_i = 0$

Moment (rotating force) can be calculated from $\vec{r} \times \vec{F}$

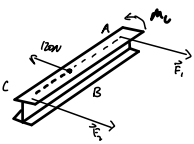
Moments positive according to right hand rule so CCW +

Reaction moments usually are caused by physical constraints

Different supports have different reaction forces/moments

Moments can be represented by FORCE COUPLES: system of two equal/equidistant forces ($M_c = F \cdot d$)

$M = \vec{r} \times \vec{F} = rF\sin\theta = \Omega F$



$\Sigma M_A = 0 = \vec{r}_1 \times \vec{F}_1 + 4m \times 120N + 6m \times \vec{F}_2 \rightarrow \vec{F}_2 = 80N$

$\Sigma M_B = -4m \times \vec{r}_1 + 0m \times \vec{F}_1 + 2m \times \vec{F}_2 \rightarrow 4\vec{F}_1 = 2\vec{F}_2 = 40N$

$\Sigma M_C = -6 \times \vec{F}_1 + -2 \times \vec{F}_2 + 0 \times \vec{F}_3$

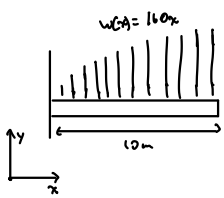
$\Sigma F_x = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 40N - 40N + 0N = 0$

Distributed Loads

Loads may not be uniformly applied to a body

In general, force can be found by integrating load wrt location $F_{total} = \int w(x)\,dx$

Can be approximated by integrating weight and applying at centroid



$F = \int_a^b w(x)\,dx = \int_0^{10} 160x\,dx$

$= 80x^2 \Big|_0^{10} = 8000N$

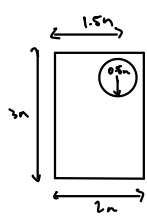
Centroids

Centroid is geometric center of shape, avg location of all points in shape

Point where total weight / distributed load acts

$\bar{x} = \frac{\Sigma(x_i A_i)}{\Sigma A_i}$ where x_i and A_i are centroid of smaller shape or hole

$\bar{x} = \frac{\int x\,dA}{A} = \frac{1}{A}\int_a^b x\,f(x)\,dx = \frac{1}{A}\int_a^b \int_c^d x\,dy\,dx$



$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{(6)(1) + (-\frac{\pi}{4})(1.5)}{6 - \frac{\pi}{4}}$

$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{(6)(1.5) + (-\frac{\pi}{4})(2.5)}{6 - \frac{\pi}{4}}$

Hydrostatic Pressure

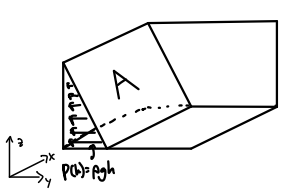
Pressure is function of depth due to weight of water above ($P(z) = \rho g z$)

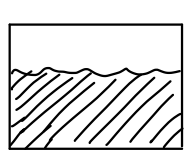
Force = Pressure × Area so hydrostatic force $= \int_{z_1}^{z_2} P(z)\,dA$

Pressure is not a function of x so we don't need to factor it in when integrating

Moment can be found using equivalent load @ centroid ($\frac{1}{3}$ of height and $\frac{1}{3}$ of width for triangles)

Distributed load applied @ centroid as $w \sim 4$





Total hydrostatic force $= \frac{1}{2} P \times L \times w = \frac{1}{2}\rho g h^2 w$

FRICTION ($F_f = \mu F_N = \mu mg$)

Two types of friction: static (object at rest), kinetic (object in motion)

Friction is a reaction force so only as strong as opposing force

Friction can cause multiple outcomes depending on force

To solve, evaluate for lowest friction and account for tipping w/ moments

Ex: do boxes slide or tip or does top slide off?

W of each = 650N

μ_s (box-box) = 0.52

μ_s (box-floor) = 0.28

F < 338N: boxes don't move

F > 338N: top slides

$0 = \sum F_{A,y} = F_N - F_G \Rightarrow F_N = F_G = 650N$

F_f is at max $0.52(650N) = 338N$

$0 = \sum F_{AB,y} = F_N - F_G \Rightarrow F_N = F_G = 2(650N)$

$F_{f,AB} = 0.28(1300N) = 364\ N$

$\sum M_P = -z \cdot F + \frac{1}{2}x(2mg)$

F > 340N, if $\sum M_P < 0$, box tips

BELT FRICTION ($T_2 = T_1 e^{\mu_s \beta}$)

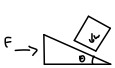
β is wrap angle in radians, usually = exit angle

T_2 is larger tension, T_1 is less tension

PULLEYS

Force exerted on larger pulley

exerted on smaller by ratio of diameter



$F\cos\theta = F_f = \mu N$

$F\sin\theta = N$

$F\cos\theta = \mu F\sin\theta$

$1 = \mu \tan(\theta)$

$\tan(\theta) > \mu$

WEDGES

Wedges can be self locking when friction forces are large enough to prevent squeeze-out

If wedge angle is greater than twice the angle of friction, wedge is self locking

$\theta > 2\tan^{-1}(\mu)$

TRUSSES

Trusses are structures composed of straight members w/ joints on end

Members are always in compression or tension w/ no bending or shear

ASSUMPTIONS: joints free to rotate w/ no reaction force

two force members: members only carry axial load

load only applied @ joints: no bending

weight of members is negligible

METHOD OF JOINTS: draw FBD of each joint and write equilibrium equations

all unknown forces should be tension

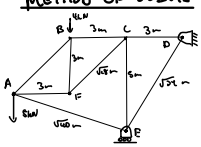
METHOD OF SECTIONS: cutting truss into sections w/ max. 3 unknowns

taking moments or force sums from either part ok

ZERO FORCE MEMBERS: joint w/ two non-collinear, non-load bearing joints

three members @ joint w/ two collinear members

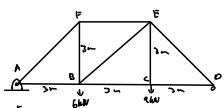
METHOD OF JOINTS

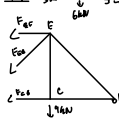


- find $\sum M_B$ and then $F_{D,ren}$
- move left and find least unknowns

METHOD OF SECTIONS

* find F_{DE}, F_{BE}, F_{EF}





- Find $\sum M_A$ for $F_{D,ren}$
- Use $\sum M_B$ of section to find F_{EF}
- Use $\sum M_E$ of section to find F_{EB}
- Use $\sum F_y$ to find F_{EB}

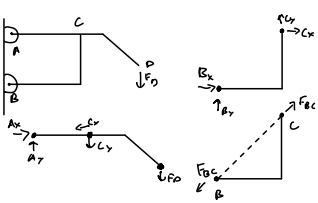
FRAMES/MACHINES

Frames are made of rigid members (some being multi-force so loaded at more than two points)

Machines are an assembly of rigid members that do work by changing input forces to output forces

All joints between parts must have equal and opposite reactions (Newton's Third Law)

Members loaded only @ two points can be treated as two-force members w/ force aligned between points



Internal Forces

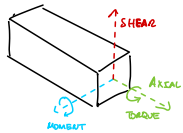
Axial Loads: loads along the beams axis

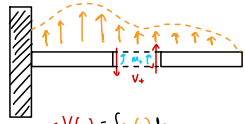
Torques: moment around axis of rotation

Moments: moment perpendicular to axis of member

Shear Forces: perpendicular load to axis

creates bending moment @ any nonzero distance





$\Delta V(x) = \int w(x)\,dx$

$\Delta M(x) = \int V(x)\,dx$

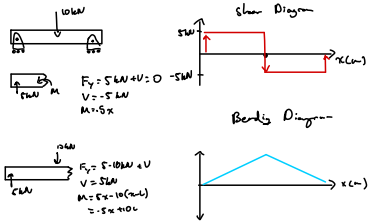
Types of External Loads

Point Loads: loads @ a specific point

Distributed Loads: loads distributed along a length of a member

External Couples: moment caused by two external forces equidistant from a point

To find max shear/bending moment, draw shear/bending moment diagrams @ critical points



Stress/Strain

Axial/Normal Stress: stress carried purely axially (as in two-force member)

Normal Stress: $\sigma = \frac{F}{A}$, $\sigma > 0$ = tension, $\sigma < 0$ = compression

Normal Strain: $\varepsilon = \frac{L^* - L}{L}$ where L^* is stretched length

Hooke's Law $\sigma = E\varepsilon$ where E = Young's modulus (stiffness of material)

$F_S = \frac{\sigma_y}{\sigma_{allow}} = \frac{\sigma_U}{\sigma_{allow}}$

$\varepsilon_y = \varepsilon_z = -\nu\varepsilon_x$ where ν = **Poisson's Ratio**: measure of deformation perpendicular to load

Shear Stress: stress perpendicular to axis

Shear Stress: $\tau = \frac{V}{A}$

Shear Strain: $\gamma = \frac{\pi}{2} - \theta = \frac{\delta_s}{L_s}$

$\therefore \frac{\pi}{2} - \theta = \frac{\delta_s}{L_s}$

Hooke's Law: $\tau = G\gamma$ where G = shear modulus

$G = \frac{E}{2(1+\nu)}$ where E = modulus of elasticity, ν = poisson's ratio

$F_S = \frac{\tau}{\tau_{allow}} = \frac{\tau}{\tau_{allow}}$

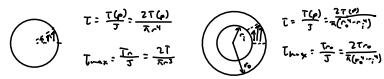
Torsional Stress: stress due to twisting

Shear @ Radius: $\tau(\rho) = \frac{T(\rho)}{J}$ where T = torque, ρ = radius, J = polar MOI of shaft cross-section

$\tau_{max} = \frac{Tr}{J}$ where $\rho = r$ (outer perimeter of shaft)

Power: $P = T\omega = T \cdot 2\pi f \rightarrow T = \frac{P}{2\pi f}$

$J = \frac{\pi}{2}\rho^4$



Normal Stress (Bending): Stress due to bending

Normal Stress: $\sigma_x(y) = \frac{-M(y)}{I}$ where M = internal moment, y = distance from neutral axis, I = second area MOI of cross section

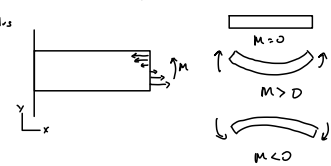
Normal Strain: $\varepsilon_x(y) = \frac{\sigma_x}{E} = \frac{-M(y)}{EI}$ where E = young's modulus

Pure Bending: $V = 0 \frac{1}{2} M \neq 0$

Bending Around Drum: $\varepsilon_x = \varepsilon_x(y) = \frac{1}{\rho}(y)$; $\sigma_x = E(\varepsilon_x) = \frac{E}{\rho}(y)$

ρ is radius − neutral distance

y is distance from "fiber" to ρ



Second Moment

$I = \int_A y^2\,dA$

Parallel Axis Theorem $I_B = I_A + Ad_{AB}^2$ where A is centroid, B is some other point

Composite Parts: determine centroid of composite, determine I for each part, apply Parallel Axis Theorem, sum all parts

$I_x = \int_A y^2\,dA$

$I_y = \int_A x^2\,dA$

$J_o = \int_A r^2\,dA = I_x + I_y$